

Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns

Robotics and Perception Engineer

Robotics and perception engineer with experience building real-time multi-camera and sensor data pipelines in Python and C++ for safety-critical environments, including systems delivered for BP and Austin Industries. Strong background in camera integration, video and sensor data processing, edge inference, and deployment reliability, with hands-on experience in multi-camera synchronisation, camera ingestion, timestamp synchronisation, and sensor calibration across ROS2 and edge systems. Experienced in troubleshooting live systems, improving throughput and latency, and producing technical documentation for deployment, maintenance, and cross-functional delivery.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Built multimodal data ingestion, cleansing, training, and validation pipelines using Python, Pandas, Apache Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours, handling upto 3.6 million datapoints.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Implemented reliability patterns and operational observability, using timeouts, retries, message validation, and structured logs and metrics to improve root-cause analysis and maintain stable behaviour under load.
- Led on-site validation and diagnostics on BP oil-rig deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.
- Documented deployment guardrails, failure modes, and recovery procedures, improving maintainability for engineers supporting fielded systems.
- Applied classical image-processing techniques to detect mud and sand splashes on camera lenses, reducing false positives by 13% and improving system reliability in harsh industrial conditions.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Built and maintained a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising camera streams and running CV-based incident detection across both feeds under edge compute constraints.
- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Implemented optical flow-based ego-motion estimation to feed motion representations into CNNs for vehicle-state classification in production pipelines.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.

- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Defined and standardised FP16/INT8 optimisation paths (including QAT, pruning, and layer fusion), reducing model size by 50% while preserving accuracy under tight memory and latency constraints.
- Worked closely with hardware, software, and QA teams to integrate perception modules into deployed systems and support rollout, optimisation, and reliability decisions.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, and RT-DETR on a shared validation subset, comparing mAP, inference time, and model size to understand trade-offs for low-latency deployment.
- Exported models to ONNX and evaluated edge deployment paths using OpenVINO, strengthening practical understanding of optimisation and model selection for production-style CV systems.
- Implemented and evaluated recent edge and foundation-model approaches, including Grounding DINO, MobileSAM, Depth Anything v2, and EfficientViT, to compare deployment trade-offs across accuracy, latency, and model size.

ROS2 SLAM & Nav2 Simulation

Feb 2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR+RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection and iteration.
- Developed ROS2 nodes, launch files, and parameterised configurations in C++ and Python, using tf2, message synchronisation, camera and sensor ingestion, and timestamp alignment to maintain frame consistency and debug transform timing issues.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation, and registration, and added an EKF-based localisation layer with robot_localization to fuse wheel odometry and IMU for smoother odom-to-base tracking during LiDAR-only SLAM.
- Containerised the stack with Docker on Linux and managed versioning in Git to support reproducible builds and consistent execution across development environments.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

AWARDS & SCHOLARSHIPS

Best Speaker award

Jan 2020

ABC Toastmasters club

SKILLS

- **Programming & Tooling:** Bash, C++17, CMake, colcon, Git, Linux, NumPy, OpenCV, Pandas, PIL, PostgreSQL, Python, scikit-image, SciPy
- **Robotics and Middleware:** Closed-loop robot control, Gazebo Harmonic, launch files, message synchronisation, Nav2, rclcpp, rclpy, Real-time perception systems, robot_localization, ROS2 (Robot Operating System 2), rosbag2, sensor data playback, Sensor integration, tf2, timestamp synchronisation
- **SLAM, Camera and Data Pipelines:** Camera ingestion, Cartographer, Ceres, Eigen, Filtering, LiDAR SLAM, LIO-SAM, Multi-camera perception, Point cloud processing, Pose graph optimisation, RTAB-Map, Scan matching, Segmentation, Video and sensor data processing pipelines, Visual data streams
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Computer Vision & Image Processing:** Data augmentation, Image processing, LiDAR extrinsics calibration, Object detection, Open3D, OpenCV, PCL (Point Cloud Library), Sensor calibration
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering
- **Computer Vision Models:** ByteTrack, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, MobileSAM, RT-DETR, SSD MobileNet V2, YOLOv10, YOLOv5, YOLOv8